Agentic AI for Clinical Decision Making and Pattern Recognition Across Treatment, Payment, and Operations

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Topic 2: Clinical Decision Making and Pattern Recognition in Health Care (TPO)

Defining the Topic

Health insurers receive millions of medical claims every year, and someone has to decide whether each one should be paid. That decision comes down to patterns in the data: the codes on the claim, the diagnosis, the amount charged, and how the provider usually bills. Clinical decision-making and pattern recognition are about using AI to find those patterns and make or support the decision.

The newest version of this is an AI agent. An agent is not like a normal program. You give it a goal and a set of tools, and it decides which tools to use, looks at the results, and reaches a decision on its own. The familiar methods still sit underneath it. Classification, prediction, clustering, and anomaly detection each answer a different question about a claim or a provider.

The same agent can work across all three sides of a health plan's business, or TPO. It can ask whether the care was right (treatment), whether the claim should be paid (payment), and whether a provider is billing normally (operations). For a company like Cotiviti, whose main job is checking whether claims are correct, this is a close fit.

Trends

A few things make this worth paying attention to now. The problem is large, and it is not going away. In 2024, CMS reported about $31.7 billion in improper Medicare fee-for-service payments, and roughly $87.1 billion across all of its programs (Centers for Medicare & Medicaid Services [CMS], 2024). Most of that is not fraud. It comes from unsupported coding or missing documentation (KFF, 2025), which is exactly the kind of thing software can catch.

Payers are also starting to act on AI. McKinsey estimates that health plans using current AI could cut administrative costs by 13 to 25 percent (McKinsey, 2024). By late 2025, more than half of surveyed healthcare organizations said they were using generative AI, and the attention has begun to shift from tools that write text to agents that take action (McKinsey, 2026a).

At the same time, governance is becoming a requirement rather than a nice-to-have. An agent can make an early mistake that carries through the rest of its work before a person ever sees it. Because of that, experts say governance has to come first (Healthcare Today, 2026). The NIST AI Risk Management Framework, built around the functions Govern, Map, Measure, and Manage, has become the standard people point to (National Institute of Standards and Technology [NIST], 2023, 2024).

Opportunities and Threats

The upside is clear. An agent can review far more claims than a manual team, and it applies the rules the same way every time. One agent can check the care, the payment, and the provider at once, instead of using three separate systems. And because it records each step, you can explain why it reached a decision. In healthcare, that explanation matters as much as the answer.

The risks are just as real, and most of them come down to trust. If the system flags the wrong claims, providers get frustrated and file appeals. If it denies something without a clear reason, that invites regulatory trouble. An agent left on its own can turn a small error into a bigger one. Patient privacy under HIPAA limits what data can be used. And leaning on a single risk score, without the exact rules that compliance depends on, is a risk by itself. The pattern is simple: power without oversight is a liability.

A Recommendation for Cotiviti

Cotiviti should test an agent-based review layer that adds machine learning and retrieval on top of its existing rules, rather than replacing them. There are three parts to this.

First, keep the rules and add reasoning. Compliance depends on exact, checkable edits. An agent should run those rules and treat a risk score or a provider signal as extra evidence, not as the reason to deny a claim on its own.

Second, treat governance as the product. Build in the human sign-off for low-confidence or high-dollar cases, keep an audit trail, and document against the NIST framework. As more of the market moves to agents, being able to show that your system is safe is a real advantage.

Third, ground policy in retrieval. Have the system pull up the actual policy behind each decision and cite it (Lewis et al., 2020). That keeps it current as policies change, and it is a step toward turning written policy into code.

The Proof of Concept and Methods

To show this can work, I built a small system called Aegis. It is an agent that reviews a claim. It plans which checks to run, calls its tools (unbundling, medical necessity, amount anomaly, duplicates, a trained risk model, provider clustering, and time-series anomaly detection), reasons over what it finds, and returns a decision approve, deny, or send to a human - inside a governance layer. On a test set of 300 labeled claims, it reached about 88 percent accuracy and caught 93 percent of the bad ones. The price checks use real CMS Medicare charge data (CMS, n.d.). The claims themselves are synthetic, so no real patient data is involved and each case is easy to reproduce. One example makes the point. The agent was 93 percent sure it should deny a high-dollar claim, but the governance layer overruled it and sent it to a person, because the amount was over the limit. That is the behavior the trends above call for.

Methods. I used public reporting from CMS, KFF, McKinsey, and NIST for the background and my own proof of concept for the feasibility. The numbers come from running the system on held-out synthetic data, and they can be reproduced from the code.

References

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